Effective Object Detection in Hyperspectral Images using Segmentation and Multibranch 2D-3D CNN Feature Fusion

Tanver Ahmed
Computer Science and Engineering
Hajee Mohammad Danesh Science and Technology University
Dinajpur, Bangladesh
tanverahmed.cse@gmail.com

Adiba Mahjabin Nitu
Computer Science and Engineering
Hajee Mohammad Danesh Science and Technology University
Dinajpur, Bangladesh
nitu.hstu@gmail.com

Masud Ibn Afjal
Computer Science and Engineering
Hajee Mohammad Danesh Science and Technology University
Dinajpur, Bangladesh
masud@hstu.ac.bd

***Abstract*—In spite of the improvement of the Hyperspectral Image classification using a convolutional neural network, lesser training samples limit the classification performance. While 3D CNNs can extract combined spectral-spatial characteristics but need more complicated models, traditional 2-D CNN approaches underutilize the interband correlations in HSIs. Combining 3D-2D CNN multibranch feature fusion can take advantage of both the 3D and 2D CNN. Our proposed method implements Segmented Principal Component analysis (SPCA), minimum-Redundancy-Maximum-Relevance (mRMR) to exploit the local structure of the data and rank the features to form groups based on their significance. To extract the features as efficiently as possible, the shallow branch carries the least significant features, and the deep branch carries the top features. Our suggested model works better than the existing ones, attaining noteworthy accuracy of 99.12%, 99.94% and 100% on Indian Pines, University of Pavia and Salinas Scene datasets respectively.**

***Index Terms*—Remote Sensing, Hyperspectral Image, CNN, Multibranch Fusion, mRMR.**

I. Introduction

A technique called hyperspectral remote sensing acquires both spectral and spatial data about a region. It generates "image cube" data, meaning that each pixel across several narrow bands has both 1D spectral information and 2D spatial features. These characteristics of the image provide information about the object's external shape along with its chemical composition and physical structure using the spectral signature. The spatial-spectral features of the Hyperspectral Image (HSI) expand the field of mineral exploration, environmental monitoring, precision agriculture, security analysis as well as food quality assurance. Deep learning has been showing promise as a large data analysis method recently, producing significant advancements across a range of computer vision jobs. As this technology has advanced, scientists have used it to classify HSI data with remarkable outcomes. The primary deeplearning is made up of convolutional neural network (CNN) [1], Stacked AutoEncoder (SAE) [2], restricted Boltzmann machine, recurrent neural network (RNN) [3], and generative adversarial network [4].

Incorporating spatial and spectral information various works have been done using deeplearning models. Exploiting 2D CNN architecture, effective spatial data extraction is done in [5]. The simultaneous processing of spatial and spectral components is accomplished via a suggested 3D CNN-based model that uses various 3D convolution sizes, leading to a reduced parameter model [6]. Incremental-PCA (IPCA) and 3D-2D CNN are used by Fast and Compact 3D-CNN (Fast 3D-CNN) for spectral-spatial feature extraction and HSI classification [7]. A wavelet-based model extracting spectral and spatial information named Spectral Network is suggested in [8] and combining 3D-2D CNN architecture Hybrid Spectral Network is proposed in [9].The CNN approach in [10] fuses the spatial information with spectral information, which are extracted separately from HSI. An artificial neural network that is led by center loss runs this process, and then the aggregated features are classified. A suggested model in [11], containing 2D CNN-based two-stream for extracting spectral and spatial features separately, fuses these outputs from the streams to produce a joint set of features including spatial and spectral information. By utilizing multi-branch feature fusion, the authors [12] use the advantages of both 2D and 3D CNN. 2D-3D CNN along with multibranch fusion and segmentation approach is shown in [13]. Their multibranch fusion improves accuracy compared to traditional CNN approaches. Segmentation techniques in Machine Learning techniques such as Segmented Principal Component Analysis (SPCA) [14], Segmented t-Stochastic Neighbor Embedding [15], Segmented Non-negative Matrix Factorization [16] show better classification accuracy. The objective of this study is to

1) Exploit the advantages of SPCA, mRMR, and Multibranch Feature Fusion.
2) Improve the classification accuracy.

The following is a summary of this article's major contributions.

1) Our innovative combination technique advances both theory and practice by combining 2D-3D CNN multibranch feature fusion, SPCA, and mRMR.
2) In addition to using theory, our approach presents a unique approach by combining information from several subregions inside HSI cubes.
3) To give empirical proof, we conduct thorough assessments on three benchmark datasets, confirming the efficacy of our method and highlighting its wide application.

This article's remaining sections are organized as follows. An

979-8-3315-2976-5/24/$31.00 ©2024 IEEE

extensive discussion of the topic's theoretical foundations is provided in section II. Section III describes the suggested neural network technique and its specific implementation procedures. In Section IV, the experiment's findings and the analysis that followed are covered in detail. In Section V, the study's results are summarized.

II. Related Theory

A. Convolutional Neural Network

Convolutional Neural Networks, or CNNs, are a type of feedforward neural network, that are well recognized for their remarkable capability to deal with demanding image processing applications [17]. Two fundamental design architectures make up the CNN. Pixels in the nearby regions are typically closely connected; considering this fact, CNN first makes use of them for the 2D structure of a picture. Secondly, CNN constructs an output feature map for each channel using a technique named feature sharing, in which the same filter is convolutionally applied to every location. The four key components construct the CNN architecture, they are- the convolutional layer, activation function, pooling layer, and fully connected layer. In the convolutional layer, computation is executed on the basis of the following equation:

$$X_i^l = \sum_{j=1}^{f} \Phi(W_{ji}^l * X_j^{l-1} + b_i^l) \tag{1}$$

where, X_j^{l-1} stands for $(l-1)$ layer's jth feature map, X_i^l refers to the lth layer's ith feature map, and f signifies the quantity of input feature maps. The bias and weight are initialized at random and denoted by b_i^l and W_{ji}^l respectively. The study utilizes a nonlinear function named rectified linear unit (ReLU), and the function Φ symbolizes it. The sign $*$ indicates the convolution process. Next is the pooling layer, which decreases the network's parameters, reduces the spatial dimension and overfitting issue. In the last section of the network, the fully connected layer joins several neurons to the preceding layer and delivers output values to the classifier. The use of backpropagation algorithm, archives the optimization of the weights and biases.

CNNs achieve their excellent ability to retrieve precise information from images by combining the four key components mentioned earlier. 3D CNN and 2D CNN are the two primary CNN-based HSI labeling techniques. HSI classification can be enhanced to a certain degree by combining 2D and 3D CNN but it is difficult to determine the ideal number of network layers since networks cannot effectively extract visual characteristics from data that is either too deep or too shallow. This study exploits the 2D CNN and 3D CNN with various kernel sizes to form a multibranch architecture, which acquires more varied and richer features.

B. Feature Fusion

In many applications, improving classification performance requires mixing information at multiple sizes. Low-level features provide higher spatial resolution and greater location-specific information. Fewer convolutional layers lead to noise and less significant feature generation. Higher-level features capture more abstract and significant information, but they are unable to distinguish finer details in the data due to their low spatial resolution. The key to enhancing model performance is to integrate the two approaches in a way that minimizes their drawbacks and maximizes their advantages [18].

Depending on whether the fusion takes place before or after prediction, fusion techniques can be grouped into early fusion and late fusion. This paper focuses on early fusion, which involves feeding features into a final predictor after they have been fused using a "concat" technique across many layers. Late fusion techniques combine the detection outputs from several layers to improve performance.

III. Proposed Neural Network Model

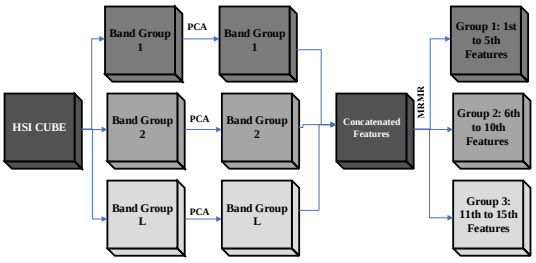


Fig. 1: Preprocessing of the HSI Datacube.

The proposed strategy integrates SPCA, mRMR, 3D-2D CNN, and multi-branch feature fusion to elevate the HSI classification performance. The SPCA and then mRMR are applied to the images for preprocessing purposes. The groups of features are made on the basis of their significance. These three groups are fed to the three-branch network to extract features at various levels. Ultimately, a softmax layer and a number of fully connected layers conduct the classification process. Fig.1 and Fig.2 illustrate the suggested paradigm.

A. Preprocessing of HSI using SPCA and mRMR

In order to address strongly correlated blocks in HSI data with low inter-correlations, SPCA is suggested as a more effective technique that may increase classification accuracy [19]. With the aim to remove spectral redundancy in local and global statistics, in this research, we first apply SPCA to HSI. Analyzing band correlations, the HSI data matrix is first split into L subgroup of datasets for SPCA implementation. Strongly connected bands are placed inside the same subgroup to reflect the dataset's local properties. The value of L in this investigation varies from 2 to 5. The previously sub-grouped data matrix sets for the SPCA implementation are indicated by the letter $\mathbf{D}_t$, and $t\varepsilon[1,2,3,4,5]$. Again, n_t is the number of bands that follow for each split data matrix. For instance, $\mathbf{D}_1 = [\mathbf{I}_1^{'}\mathbf{I}_2^{'}...\mathbf{I}_{n_1}^{'}]$, $\mathbf{D}_2 = [\mathbf{I}_2^{'}\mathbf{I}_2^{'}...\mathbf{I}_{n_2}^{'}]$ and so on. Here, the first n_1 rows of perspective $\mathbf{I}_j$ make up the $\mathbf{I}_j^{'}$. Similarly, skipping those first n_1 rows, the next n_2 rows from $\mathbf{I}_j$ make up $\mathbf{I}_j^{``}$. These procedures are carried out for each $j, n\varepsilon[1,S]$, where the spatial dimension is signified as $S = X \times Y$. The projection matrices of each subgroup's D_t are successively combined to yield the overall projection matrix. Let's consider the whole input data matrix $\mathbf{D} = X \times Y \times F$, where F stands for the spectral bands. The data matrix $\mathbf{D}$ is converted to $\mathbf{Z}_t = X \times Y \times B_t$, where B_t is the sum of features extracted from t segments.

The next step of the preprocessing task is to apply mRMR to $\mathbf{Z}_t$ [20]. By reducing duplication among the characteristics that are chosen, mRMR seeks to identify a subset of features

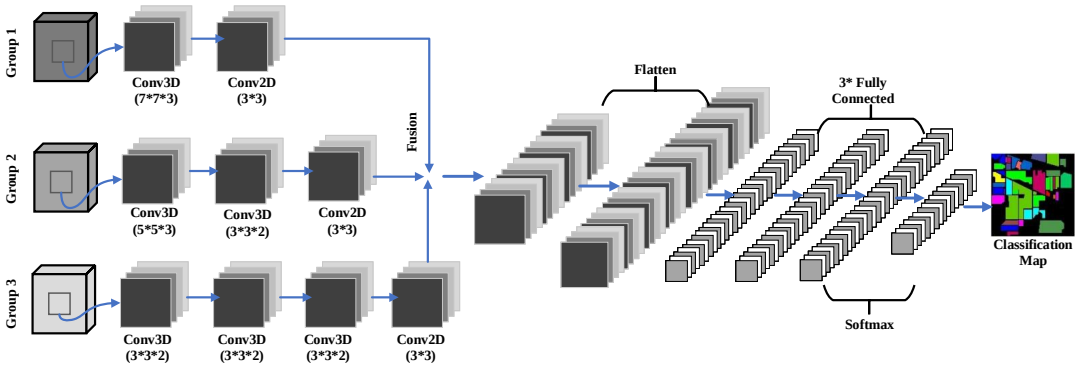


Fig. 2: Proposed Multibranch structure.

TABLE I: Segmentation details of the Datasets

Segment no (L)	Parameters	IP			SA			PU	
		Seg-3	Seg-4	seg-5	Seg-2	Seg-3	Seg-4	Seg-2	Seg-3
1	Band Range	1 to 36	1 to 36	1 to 36	1 to 36	1 to 36	1 to 36	1 to 37	1 to 37
	Band Count	36	36	36	36	36	36	37	
	Average Correlation	0.85	0.85	0.85	0.91	0.91	0.91	0.96	
2	Band Range	37 to 102	37 to 80	37 to 79	37 to 204	37 to 106	37 to 81	38 to 103	38 to 72
	Band Count	66	44	43	168	70	45	66	35
	Average Correlation	0.58	0.88	0.89	0.34	0.83	0.9	0.58	0.96
3	Band Range	103 to 200	81 to 102	79 to 102	N/A	107 to 204	82 to 106	N/A	73 to 103
	Band Count	98	22	23		98	25		31
	Average Correlation	0.95	0.5	0.51		0.96	0.94		0.95
4	Band Range	N/A	103 to 200	103 to 162	N/A	N/A	107 to 204	N/A	N/A
	Band Count		98	60			98		
	Average Correlation		0.95	0.95			0.96		
5	Band Range	N/A	N/A	163 to 200	N/A	N/A		N/A	N/A
	Band Count			38					
	Average Correlation			0.96					

that are extremely pertinent to the target class. Utilizing the mutual information among the features denoted as M, the feature relevance and redundancy are formulated as follows:

$$Relevance, R_e = \frac{1}{|B_t|} \sum_{b_i \varepsilon B_t} M(b_i; c) \tag{2}$$

$$Redundancy, R_d = \frac{1}{|B_t|^2} \sum_{b_i, b_j \varepsilon B_t} M(b_i, b_j) \tag{3}$$

where B_t is the feature set obtained after applying SPCA, b_i is the individual feature, and c is the sample class. By increasing relevance and avoiding repetition at the same time, the mRMR ranks features. And the process is implemented using an operator ϕ as follows-

$$Max\phi(Relevance, Redundancy) = (R_e - R_d) \tag{4}$$

After applying the mRMR, we select the top 15 features and split them into three different groups. The first group, which represented the highest priority tier, was made up of the top five features. The second and third groups consist of 5 features, excluding first 5 and 10 features respectively, on the basis of significance. The resultant groups are denoted by $\mathbf{Z}_t$, and $t \varepsilon [1,2,3]$.

To run our multi-branch 3D-2D CNN model, the HSI data cube is broken down into small overlapping 3D patches. The patches indicated by $P_t = s \times s \times B_t$ built from the 3 different reduced datacubes $\mathbf{Z}_t$ separately focusing on centered spatial location (i, j). Every patch has a geographical coverage area of $s \times s$ contains all the spectral bands of respective group. The quantity of the created 3D patches represented as n computed as $(X - s + 1) \times (Y - s + 1)$ where X and Y symbolizes the spatial domain of the data cube.

B. Multibranch CNN

Because of their strongly linked bands, HSI requires a lot of training samples in order to be accurately classified. However, because of the high expense and complexity needed, getting large amounts of ground reference data is difficult. Insufficient training data might result in overfitting and unpredictable model parameters. Shallow neural network can bring out only the basic edge information from images and can represent simple functions. In contrast, having deep non-linear structure, deep neural networks can extrapolate complex functions, discover distributed data representations, and show a strong capacity for learning to describe datasets using a small number of training examples. 2D CNN ignores the rich interband information content of HSIs while extracting spatial information. Overusing of 3D CNN makes the model too heavy, and may potentially compromise classification accuracy. To alleviate this issue, we combined 2D CNN and 3D CNN to develop multibrunch feature fusion model in this study. We have established three branches with different numbers of layers in the branches using the combination of 2D and 3D CNN

TABLE II: Configuration of Deep Neural Network Used in Feature Learning Procedure

Branch	Layer	Input Shape	Output Shape	Kernel-size	Filters	Connected to
	Input_1	–	(25,25,5,1)	–	–	–
	Conv3D_1_1	(25,25,5,1)	(19,19,3,8)	7×7×3	8	Input_1
1	Reshape_1	(19,19,3,8)	(19,19,24)	–	–	Conv3D_1_1
	Conv2D_1	(19,19,24)	(17,17,64)	3×3	64	Reshape_1
	MaxPool_1	(17,17,64)	(8,8,64)	2×2	–	Conv2D_1
	Input_2	–	(25,25,5,1)	–	–	
	Conv3D_2_1	(25,25,5,1)	(21,21,3,8)	5×5×3	8	Input_2
2	Conv3D_2_2	(21,21,3,8)	(19,19,2,16)	3×3×2	16	Conv3D_2_1
	Reshape_2	(19,19,2,16)	(19,19,32)	-	-	Conv3D_2_2
	Conv2D_2	(19,19,32)	(17,17,64)	3×3	64	Reshape_2
	MaxPool_2	(17,17,64)	(8,8,64)	2×2	–	Conv2D_2
	Input_3	–	(25,25,5,1)	–	–	
	Conv3D_3_1	(25,25,5,1)	(23,23,4,8)	3×3×2	8	Input_3
3	Conv3D_3_2	(23,23,4,8)	(21,21,3,16)	3×3×2	16	Conv3D_3_1
	Conv3D_3_3	(21,21,3,16)	(19,19,2,32)	3×3×2	32	Conv3D_3_2
	Reshape_3	(19,19,2,32)	(19,19,64)	-	-	Conv3D_3_3
	Conv2D_3	(19,19,64)	(17,17,64)	3×3	64	Reshape_3
	MaxPool_3	(17,17,64)	(8,8,64)	2×2	–	Conv2D_3

architecture. Layers in the branches contain different kernel sizes; ReLU is used as the activation function. Table II provides a full description of the parameters and distribution of every layer in the suggested model. Every branch is composed of one or multiple 3D CNN layers, one 2D CNN layer having 64 filters of 3×3 size kernel and one MaxPool layer of size 2×2. The very first branch has 8 3D filters of size $7 \times 7 \times 3$followed by one 2D CNN and one MaxPool layer. The next branch comes with 2 layers of 3D CNN. The first one has 8 3D filters of size $5 \times 5 \times 3$, and the other has 16 filters of size $3 \times 3 \times 2$. At last, the branch contains the only 2D CNN layer and the only MaxPool layer. Obtaining deeper semantic characteristics is the aim of the third branch. This branch is made up of three 3D CNN layers of the same kernel size of $3 \times 3 \times 2$ and the number of the filters is 8, 16, and 32 respectively. Next to these layers are the 2D CNN layer and MaxPool layer. The lengths, widths, and depths of the feature cubes that were taken across the three network branches are all the same. These three properties are merged to form the image's combined feature by taking the image's width and length into account.

IV. Experimental Result and Analysis

A. Description of the dataset and experimental setup

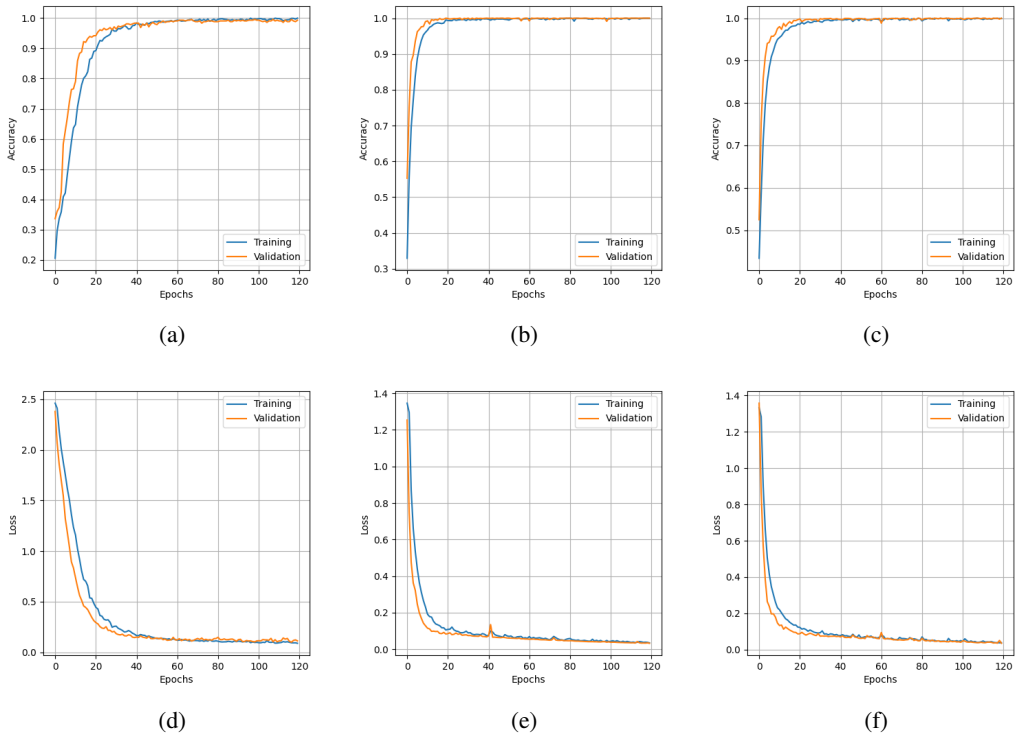
We conducted tests with the Indian Pines (IP), University of Pavia (PU), and Salinas Scene (SA) HSI datasets, all of which are publicly accessible. Images in IP have a dimension of 145 pixels wide by 145 pixels and 145 pixels high and are split into 224 distinct spectral bands that span wavelengths between 0.4 and 2.5 μ meters. Of those spectral bands, we eliminated 24 bands because they are associated with water absorption. There are sixteen distinct kinds of vegetation seen in the images, according to the ground truth data for the IP dataset. On the other hand, the images of PU have a spatial dimension of 610 $\times$ 340 pixels with 103 distinct bands spanning the wavelength ranging from 0.43 to 0.86 μ meters. The dataset contains 9 different classes. The third dataset SA, includes images of 512 $\times$ 217 spatial orientation. Similar to IP, SA has 16 distinct classes and 224 spectral bands among them, 20 are discarded because of the water absorption. Because of the diversity in spatial resolution, the quantity of spectral bands and the land cover, we chose these three datasets.

We conducted our experiments on Kaggle GPU P100. Based on a performance evaluation of the classification model under various learning rate settings, we arrived at the ideal value of 0.0001 for the learning rate hyperparameter. The network is trained without batch normalization or data augmentation using mini-batches of size 64 for 120 epochs for extracting the detailed information. The window size for the pixel was set to 25 $\times$ 25. The optimal value of L2 of l2 regularizer was set to 0.0001. The output layer includes the SoftMax activation function, and everywhere else we used ReLU activation function. The dropout rate in the fully connected layers is 40

All three datasets were split into training and testing sets. We conducted our experiments splitting the datasets two times. In the first time, we kept 10% of the data for training purposes and 90% of testing. The next experiment went on 20% training dataset and 80% testing dataset. 20% of the training data was kept for validation during the training period. Six widely used numerical measurements are used to assess the performance of the classification model's predictions in order to examine the results. They are- Overall Accuracy (OA), Kappa Coefficient (Kappa), Average Accuracy (AA), Precision, Recall and F1 Score.

B. Findings of the experiments

We considered 15 extracted features, or principal components, for all the state-of-the-art approaches and the proposed network. The proposed architecture is based on the segmentation approach. The segmentation of the bands was made on the basis of band-to-band correlation as mentioned in Table I. Because of the segmentation approach, our proposed model has some variations. The outcomes of all the variations are listed in Table III. For example, Seg-3 means, three groups of bands were made before applying PCA. Table III also compares the variations. Among them, the best-performing models are indicated in bold. Of all the variants, for 20% training data, Seg-5, Seg-3 and Seg-2 outperform the other variants in terms of the considered measurement matrices for IP, SA and PU datasets respectively. Outcomes remain the same for the 10% training dataset except for the PU dataset. Seg-3 performs better here.



(a) (b) (c)

(d) (e) (f)

Fig. 3: Accuracy Curve for a) IP, b) SA c) PU and loss curve d) IP e) SA f) PU for 20% training datasets

TABLE III: Results of the proposed model

Dataset	Number of Segments	Training Samples	OA (%)	AA (%)	Kappa (%)	Precision	Recall	F1 Score
Indian Pines	Seg-3	10%	97.02	90.69	96.60	96.85	97.02	96.89
		20%	99.11	99.28	98.99	99.12	99.11	99.11
	Seg-4	10%	96.95	**93.46**	96.53	96.96	96.95	96.93
		20%	99.11	96.42	98.98	99.11	99.11	99.10
	Seg-5	10%	**97.40**	90.79	**97.03**	**97.26**	**97.40**	**97.30**
		20%	**99.12**	**99.35**	**99.00**	**99.15**	**99.12**	**99.13**
Salinas	Seg-2	10%	99.93	99.89	99.92	99.93	99.93	99.93
		20%	99.98	99.98	99.98	99.98	99.98	99.98
	Seg-3	10%	**99.97**	**99.97**	**99.97**	**99.97**	**99.97**	**99.97**
		20%	**100**	**100**	**99.99**	**100**	**100**	**100**
	Seg-4	10%	99.96	99.96	99.95	99.96	99.96	99.96
		20%	99.99	99.99	99.99	99.99	99.99	99.99
Pavia University	Seg-2	10%	99.36	98.49	99.15	99.36	99.36	99.36
		20%	**99.94**	**99.93**	**99.93**	**99.95**	**99.95**	**99.94**
	Seg-3	10%	**99.49**	**98.83**	**99.33**	**99.49**	**99.49**	**99.49**
		20%	99.93	99.89	99.91	99.93	99.93	99.93

The comparison with the state-of-the-art is illustrated in TableIV. Seg-5 for IP, Seg-3 for SA outperform all the state-of-arts for both the 10% and 20% training samples. Seg-3 and Seg-2 outperform all the state-of-arts for 10% and 20% training samples respectively. UP contains fewer categories than the other three datasets—only nine—making it simpler to categorize. Because SA contains the greatest number of spectral bands and a big spatial extent, the stated approaches perform better in terms of classification on SA. Despite having 16 categories, IP has a very limited spatial size, which lowers classification accuracy.

In our approach, after applying SPCA, we determine the significance of the bands using mRMR feature selection technique. Three groups were made using mRMR. The group containing the least significant features is carried to the shallow network branch of the model. The most noteworthy features are passed to the deep branch of the model and the other one is passed to the second branch. The retrieved features achieve tiny intraclass differences and big interclass differences, and the network learns the discriminative and representative features by fusing them together. We covered the local structure of the dataset using the segmented approach and made the most significant to least significant groups and processed them to deep to shallow feature extraction. Our suggested approach increases the OA by 1.68%, 0.38% and 0.74% when compared to the best method, HybridSN for IP, SA, PU respectively with less features. The accuracy curve and loss curve can be illustrated from fig 3.

V. Conclusion

In this study, we propose a model that integrates two-step preprocessing techniques SPCA and mRMR. The preprocessing technique builds three different groups of features according to their significance and on the basis of this significance, these

TABLE IV: Comparison of the proposed and state-of-the-art techniques

Dataset	Model	Trainig Samples	OA (%)	AA (%)	Kappa (%)	Precison	Recall	F1 Score
Indian Pines	2D CNN [5]	10%	42.33	54.79	36.63	53.36	51.35	52.34
		20%	53.54	71.65	51.25	52.25	53.30	52.79
	Fast Compact 3D CNN [7]	10%	71.11	51.28	66.54	69.67	71.11	68.64
		20%	83.39	61.03	80.95	81.54	83.38	82.03
	Hybrid SN [9]	10%	95.72	89.25	95.11	95.64	95.71	95.56
		20%	99.09	98.35	98.96	99.11	99.08	99.09
	2D-3D Multibranch [12]	10%	93.08	86.90	92.08	93.01	93.08	92.91
		20%	98.67	98.35	98.48	0.9870	98.66	98.66
	Proposed	**10%**	**97.4**	**90.79**	**97.03**	**97.26**	**97.40**	**97.30**
		20%	**99.12**	**99.35**	**99.00**	**99.15**	**99.12**	**99.13**
Salinas Scene	2D CNN [5]	10%	80.02	93.67	78.20	79.82	78.88	79.39
		20%	85.65	95.52	84.19	84.91	85.91	85.41
	Fast Compact 3D CNN [7]	10%	99.09	99.54	98.99	99.09	99.08	99.09
		20%	98.76	97.35	98.62	98.91	98.76	98.73
	Hybrid SN [9]	10%	99.59	99.76	99.54	99.59	99.58	99.50
		20%	99.97	99.93	99.97	99.97	99.97	99.97
	2D-3D Multibranch [12]	10%	99.88	99.91	99.86	99.87	99.87	99.87
		20%	99.97	99.96	99.97	99.96	99.97	99.97
	Proposed	**10%**	**99.97**	**99.97**	**99.97**	**99.97**	**99.97**	**99.97**
		20%	**100**	**100**	**99.99**	**100**	**100**	**100**
Pavia University	2D CNN [5]	10%	89.71	88.81	86.47	88.88	89.50	89.19
		20%	94.46	94.68	92.71	93.39	92.42	92.90
	Fast Compact 3D CNN [7]	10%	97.68	97.24	96.92	97.88	97.67	97.69
		20%	99.46	99.12	99.28	99.45	99.45	99.45
	Hybrid SN [9]	10%	98.75	98.79	98.35	98.80	98.74	98.75
		20%	99.88	99.76	99.84	99.87	99.87	99.87
	2D-3D Multibranch [12]	10%	99.07	98.62	98.77	99.09	99.73	99.00
		20%	99.62	99.54	99.49	99.62	99.61	99.61
	Proposed	**10%**	**99.49**	**98.83**	**99.33**	**99.49**	**99.49**	**99.49**
		20%	**99.94**	**99.93**	**99.93**	**99.95**	**99.95**	**99.94**

groups are carried out to the combination of 2D-3D CNN multibranch network. The suggested technique creates a multi-branch feature fusion framework and optimizes the CNN-based model's development. The experimental findings demonstrate that the suggested strategy performs well in terms of OA, AA, Kappa, recall, precision, and f1 score. The IP dataset has slightly inferior performance to the other two datasets when using the suggested technique. Therefore, our future work will focus on investigating a more condensed neural network model, enhancing generalization capability, and achieving adequate classification accuracy on a variety of datasets.

REFERENCES

[1] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, 2017. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231216310104

[2] F. Lv, M. Han, and T. Qiu, "Remote sensing image classification based on ensemble extreme learning machine with stacked autoencoder," *IEEE Access*, vol. 5, pp. 9021–9031, 2017.

[3] H. Wu and S. Prasad, "Convolutional recurrent neural networks forhyperspectral data classification," *Remote Sensing*, vol. 9, no. 3, 2017. [Online]. Available: https://www.mdpi.com/2072-4292/9/3/298

[4] X. Wang, K. Tan, Q. Du, Y. Chen, and P. Du, "Caps-triplegan: Gan-assisted capsnet for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 9, pp. 7232–7245, 2019.

[5] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2015, pp. 4959–4962.

[6] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-d deep learning approach for remote sensing image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 56, no. 8, pp. 4420–4434, 2018.

[7] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-d cnn for hyperspectral image classification," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.

[8] T. Chakraborty and U. Trehan, "Spectralnet: Exploring spatial-spectral waveletcnn for hyperspectral image classification," 04 2021.

[9] S. Roy, G. Krishna, S. R. Dubey, and B. Chaudhuri, "Hybridsn: Exploring 3d-2d cnn feature hierarchy for hyperspectral image classification," 02 2019.

[10] A. J. X. Guo and F. Zhu, "A cnn-based spatial feature fusion algorithm for hyperspectral imagery classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 9, pp. 7170–7181, 2019.

[11] X. Li, M. Ding, and A. Pižurica, "Deep feature fusion via two-stream convolutional neural network for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 4, pp. 2615–2629, 2020.

[12] Z. Ge, G. Cao, X. Li, and P. Fu, "Hyperspectral image classification method based on 2d–3d cnn and multibranch feature fusion," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 5776–5788, 2020.

[13] M. I. Afjal, M. Mondal, and M. A. Mamun, "Effective hyperspectral image classification based on segmented pca and 3d-2d cnn leveraging multibranch feature fusion," *Journal of Spatial Science*, pp. 1–28, 01 2024.

[14] X. Jia and J. Richards, "Segmented principal components transformation for efficient hyperspectral remote-sensing image display and classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 37, no. 1, pp. 538–542, 1999.

[15] T. Ahmed, M. Bari, M. I. Afjal, A. Nitu, M. P. Uddin, and M. Marjan, *Effective Feature Extraction via Segmented t-Stochastic Neighbor Embedding for Hyperspectral Image Classification*, 05 2023, pp. 37–48.

[16] M. H. Bari, T. Ahmed, M. I. Afjal, A. M. Nitu, M. P. Uddin, and M. A. Marjan, "Segmented nonnegative matrix factorization for hyperspectral image classification," in *2023 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2023, pp. 1–5.

[17] H.-J. Yoo, "Deep convolution neural networks in computer vision: a review," *IEIE Transactions on Smart Processing and Computing*, vol. 4, pp. 35–43, 02 2015.

[18] S. Chaib, H. Liu, Y. Gu, and H. Yao, "Deep feature fusion for vhr remote sensing scene classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 8, pp. 4775–4784, 2017.

[19] M. P. Uddin, M. A. Mamun, M. I. Afjal, and M. A. Hossain, "Information-theoretic feature selection with segmentation-based folded principal component analysis (pca) for hyperspectral image classification," *International Journal of Remote Sensing*, vol. 42, no. 1, pp. 286–321, 2021. [Online]. Available: https://doi.org/10.1080/01431161.2020.1807650

[20] Y. Jiang and C. Li, "mrmr-based feature selection for classification of cotton foreign matter using hyperspectral imaging," *Computers and Electronics in Agriculture*, vol. 119, pp. 191–200, 2015. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0168169915003300